

DETAILED DESCRIPTION OF THE INVENTION — DISPOSABLE, REUSABLE, AND CLINICAL DEPLOYMENT CONFIGURATIONS

The intraluminal physiologic monitoring device may be configured for disposable, reusable, semi-disposable, or hybrid deployment models depending on intended use environment, regulatory classification, hygiene requirements, or economic considerations.

In disposable embodiments, the entire device may be designed for single-use operation followed by safe disposal. Disposable configurations may reduce sterilization requirements, minimize cross-contamination risk, and simplify deployment in clinical or consumer environments. Low-cost materials, integrated electronics, or limited-duration power systems may be optimized for such applications.

Reusable embodiments may incorporate durable materials, rechargeable power systems, and sealed electronics capable of repeated insertion cycles. Reusable devices may support cleaning, sterilization, or disinfection procedures using chemical agents, ultraviolet exposure, thermal treatment, or other approved sanitation methods.

Hybrid configurations may separate reusable electronic cores from disposable sensing interfaces. In such implementations, user-contacting components may be replaced between uses while processing electronics remain protected within reusable modules. Hybrid architectures may balance hygiene, sustainability, and cost efficiency.

Clinical deployment configurations may include sterile packaging, validated sterilization compatibility, traceability labeling, clinician-guided calibration workflows, or integration with hospital information systems. Devices may be adapted for diagnostic, monitoring, research, or procedural support applications.

Consumer deployment embodiments may prioritize ease of use, automated calibration, guided insertion workflows, and privacy-preserving data presentation. Packaging, accessories, and companion software may be optimized for non-clinical environments while maintaining measurement reliability.

Certain embodiments may support multi-patient clinical reuse through replaceable sterile covers, barrier systems, or detachable sensing cartridges. Tracking mechanisms may record usage cycles, sanitation status, or operational history to ensure safety compliance.

Manufacturing and regulatory pathways may vary between deployment configurations without departing from the underlying invention. Devices may be classified across wellness, medical monitoring, diagnostic, or research categories depending on implementation and jurisdictional requirements.

All disposable, reusable, and hybrid deployment strategies capable of enabling intraluminal physiologic monitoring are considered within the scope of the present disclosure.